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Unilateral true hermaphrodite with 46,XX/46,XY dispermic chimerism.

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Abstract

A 13 year old female presented with ambiguous external genitalia, right inguinal ovotestis, left ovary, apparently normal Mullerian system, and absent Wolffian system. Cultured lymphocytes showed a 46,XX/46,XY karyotype. Histopathology of the gonads confirmed true hermaphroditism. The presence of two genetically different erythrocyte populations was observed. The findings suggested that the patient is a true hermaphrodite dispermic chimera.

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SUMMARY A 13 year old female presented with ambiguous external genitalia, right inguinal ovotestis, left ovary, apparently normal Mullerian system, and absent Wolffian system. Cultured lymphocytes showed a 46,XX/46,XY karyotype. Histopathology of the gonads confirmed true hermaphroditism. The presence of two genetically different erythrocyte populations was observed. The findings suggested that the patient is a true hermaphrodite dispermic chimera.

In true hermaphroditism both ovarian tissue, including oocytes, and testicular tissue, including germinal cells or tubules, are present in the same subject. This differs from mixed gonadal dysgenesis with unilateral testis and contralateral dysgenetic ovarian 'streak' tissue or agonadia.[1–7] This is the first well documented report of a true hermaphrodite dispermic chimera in the Arab population.

True hermaphrodites have been represented in art and literature for centuries, but Klebs is usually credited with the first scientific report.[8] In 1959, the first chromosomal abnormality was reported.[9 10] The first dispermic chimera true hermaphrodite with 46,XX/46,XY karyotype and two erythrocyte populations was described in 1962.[11] Almost all XX/XY chimeras are true hermaphrodites, but not all true hermaphrodites are chimeras.[6]

Case report

The proband was born on 15.8.72 after a term pregnancy to young, consanguineous, Arab parents. The grandparents of the proband were also consanguineous. Ambiguity of the external genitalia was noted by the obstetrician at birth. The child was reared as a female.

At seven years, she was a pleasant, well developed, normal young female. Genital abnormalities included a prominent phallus and glans with a terminal non-functioning urethral orifice. A separate functioning urethral opening and a small vaginal orifice were present in the vestibule. A small, tender, right sided inguinal swelling was palpable. Both serum electrolytes and urinary 17 ketosteroids were normal.

Metaphase chromosomes were prepared from peripheral blood culture. Both G and Q banding was performed. Karyotype analysis of 100 metaphase spreads revealed two distinct cell lines, 46,XX and 46,XY, in a ratio of 80:20.

Laparotomy showed a left ovary, normal Mullerian system, and absent Wolffian system. A tender right inguinal swelling was removed and a wedge biopsy from the left ovary was taken. Corrective surgery was performed to restore the normal female external genitalia by excising more than an inch of the phallus; clitoroplasty and vaginoplasty were also performed.

Histopathological study of the excised inguinal swelling confirmed the presence of ovotesticular tissue, with a spiral furrow in the middle separating it into two modular areas. One showed testicular elements with infantile tubules, and the second showed ovarian tissue with signs of oogenesis, including primordial, growing, atretic, and cystic follicles (figure). Left gonadal biopsy showed primordial ovarian follicles.

At 12 years she was comprehensively reassessed. Her hormonal profile showed a normal prepubertal female pattern and response. Psychological and psychiatric assessment confirmed her to be a well balanced female of average intelligence and feminine gender identity.

The proband and her father were typed for blood groups ABO, Rh, MNSs, P, Lu, Le, Fy, Jk, Xg, and Co. The mother's blood was not available for study. Two erythrocyte populations were detected by a mixed field agglutination picture when the proband's cells were tested with anti-M, anti-C, and anti-K. The two populations of erythrocytes were separated using anti-M. About 20% of her red cells were agglutinated using anti-M: after separation the agglutinates were 'deagglutinated' by treatment with DTT (dithiothreitol), and both 'deagglutinated' and free cells were phenotyped (table). The major

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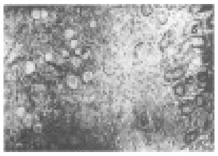

Image
on p.785

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